

- (3) $\exists$ integer n s.t. $2n^2 - 5n + 2$ is prime.
integer n is prime if $n > 1$ and
 $\forall$ positive integers r and s , $n = rs$ if
($r = n$ and $s = 1$) or ($r = 1$ and $s = n$)

* proof by example:
attempt

$$3 = 2n^2 - 5n + 2$$

$$1 = 2n^2 - 5n$$

$$1 = 2n(n - \frac{5}{2}n)$$

* attempt 2 $n=2$: $2(2)^2 - 5(2) + 2 = 0$, Not prime

$n=3$: $2(3)^2 - 5(3) + 2 = 5$, prime!

$n=3$ satisfies the original statement

- (4) disprove: $\forall$ integers m and n , if $2m+n$ is odd
 m and n are both odd

$$m=2, n=1: 2(2)+1=5$$

Original statement is wrong since $m=2$ is even, but the outcome is odd

- (5) The average of any two ints is odd.
avg of 3 and 5 is 4: even

avg of 7 and 3 is 5: odd

this statement can be true for
some odd integers but false for others

- (6) prove that the sum of any two odd integers is even
suppose that n and m are integers

$$s = (2n+1) + (2m+1)$$

$$s = 2n + 2m + 2 \rightarrow s = 2(n+m) + 2$$

$(n+m)$ is an integer since the set of
integers is closed under addition

so by the definition of odd

numbers, s (the sum) is an odd integer

(7) m is an odd integer, $n = (2m+1)$
 $n^2 = (2m+1)^2 \rightarrow n^2 = 4m^2 + 4m + 1$
 $n^2 = 2(2m^2 + 2m) + 1$

$(2m^2 + 2m)$ is an integer since the set of integers is closed under addition and multiplication, so $2(2m^2 + 2m) + 1$ is odd. by the definition of odd numbers, n^2 is odd

(8) the set of integers is closed under division, disprove by example: $8/3 = 2.\overline{666}$, which is not an integer

(9) odd integer $m = 2k+1$
 even integer $n = 2t$
 difference $= 2k+1 - 2t$
 $= 2(k-t) + 1$

Since k and t are integers (closed under subtraction), the difference is odd

(10) $a = 2k+1$, $b = 2t+1$
 $a^2 + b^2 = (2k+1)(2k+1) + (2t+1)(2t+1)$
 $= 4k^2 + 4k + 1 + 4t^2 + 4t + 1$
 $= 2(2k^2 + 2k + 2t^2 + 2t) + 2$

$(2k^2 + 2k + 2t^2 + 2t)$ is an integer since closed under addition and multiplication, so $2(2k^2 + 2k + 2t^2 + 2t)$ is even, and adding 2 to an even number is even, so by definition of an even number, $a^2 + b^2$ is even

Maya Gera HOMEWORK 3

- 1)
- a) 4 is even because dividing 4 by 2 gives an integer.
 - b) 0 is even because $0/2$ is 0 (which has a remainder of 0). $2 \cdot 0 = 0$
 - c) $6a^2b^2$ is even since a and b are even and by definition an even number times an even number gives an even number, so $a \cdot a$ and $b \cdot b$ is even, so a^2b^2 is therefore even. 6 is also an even number so $6 \cdot a^2b^2$ is even.
 - d) $-5/2$ does not give an even number and has a remainder of 1 so by definition it is odd. $-5 = (2)(-2) + (-1)$

2) 3 distinct ints m & n s.t. $\frac{1}{m} + \frac{1}{n}$ is an integer

attempt 1: $\frac{1}{m} + \frac{1}{n} = \frac{m+n}{mn}$. $m+n = k$, where k is an integer due to the fact that the set of integers is closed over addition and m & n are integers. also $mn = t$ where t is an integer since integers are closed over multiplication.

*this proof was a failed attempt since $\frac{k}{t}$ doesn't necessarily result in an integer

attempt 2: Proof by counterexample:

$\nexists$ distinct ints m & n s.t. $\frac{1}{m} + \frac{1}{n}$ is not an integer.
 $\frac{1}{m} + \frac{1}{n} = \frac{m+n}{mn}$. $m+n = k$ where k is an integer since integers closed under addition.
 $mn = t$ where t is an integer since closed under multiplication. $\frac{k}{t}$ is not necessarily an integer because the set of integers are not closed under division. therefore $\frac{1}{m} + \frac{1}{n}$ is not an integer.